

MASSIF

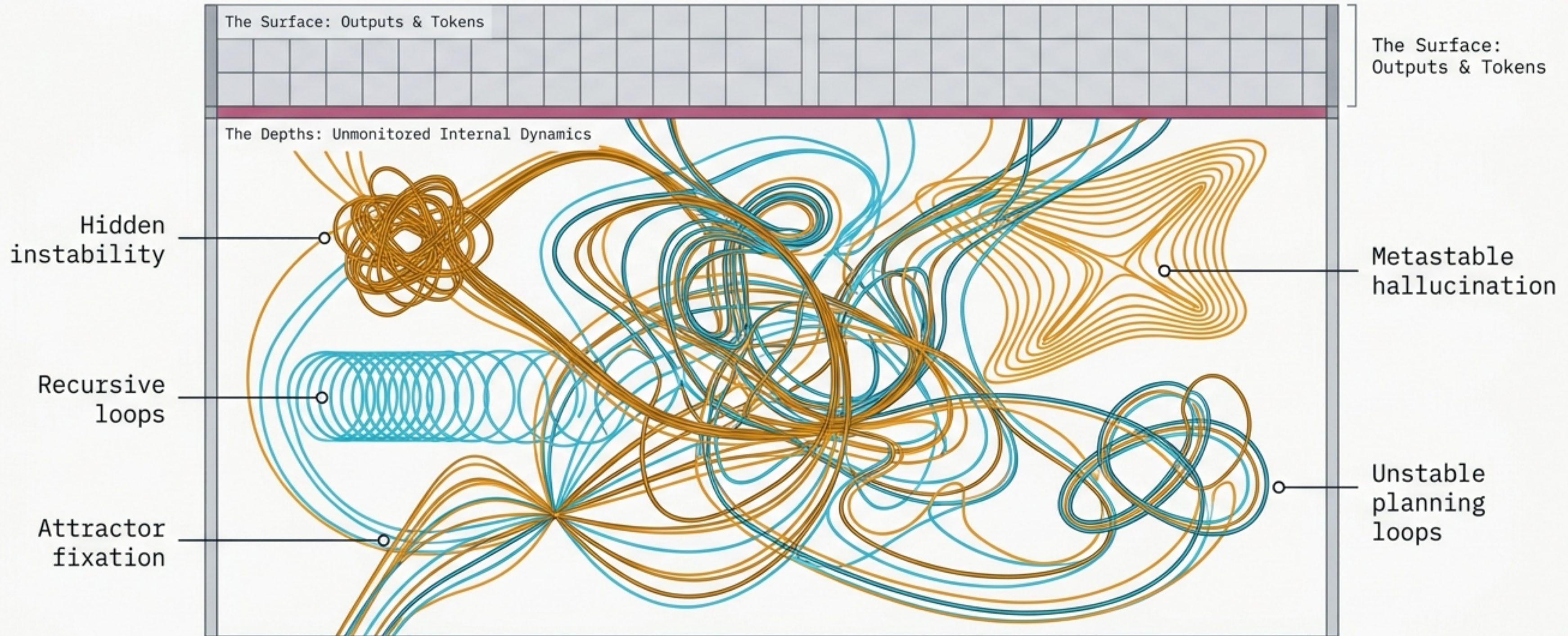
Recursive Entrainment and Latent Synchronization
in Autoregressive Language Models



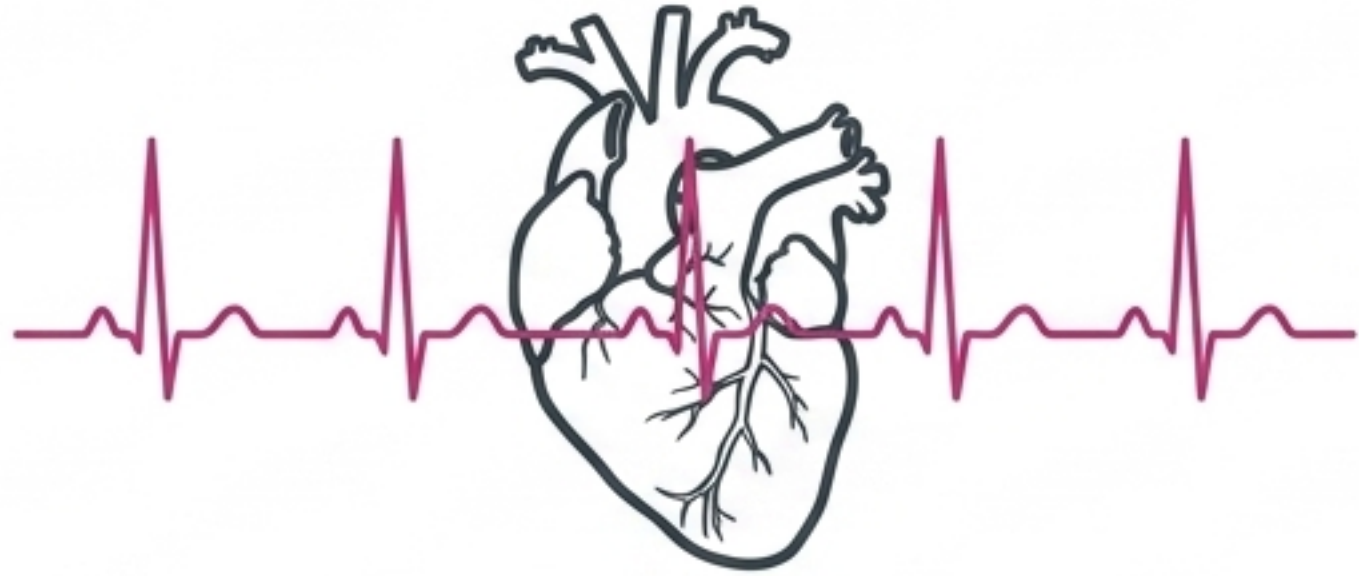
Computational Latent Dynamics
Real-Time Stability Telemetry

Daniel Solis
Principal Investigator

Current AI safety mostly studies outputs. But advanced systems may fail internally before they fail behaviorally.



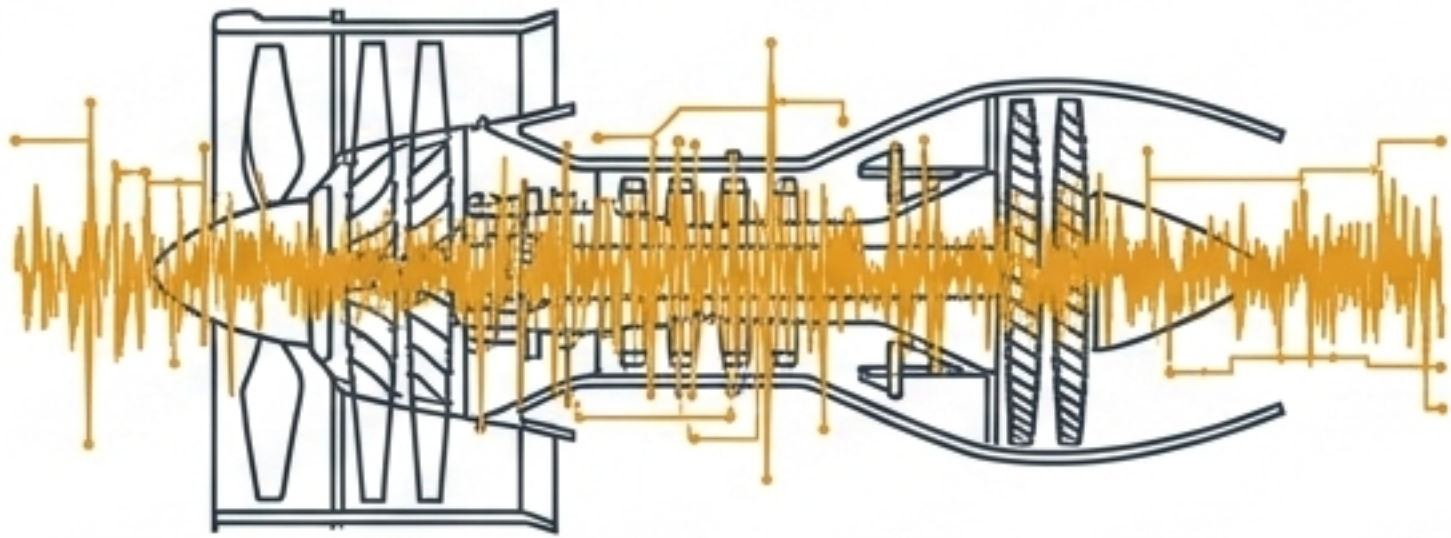
We need physiological telemetry for artificial cognition.



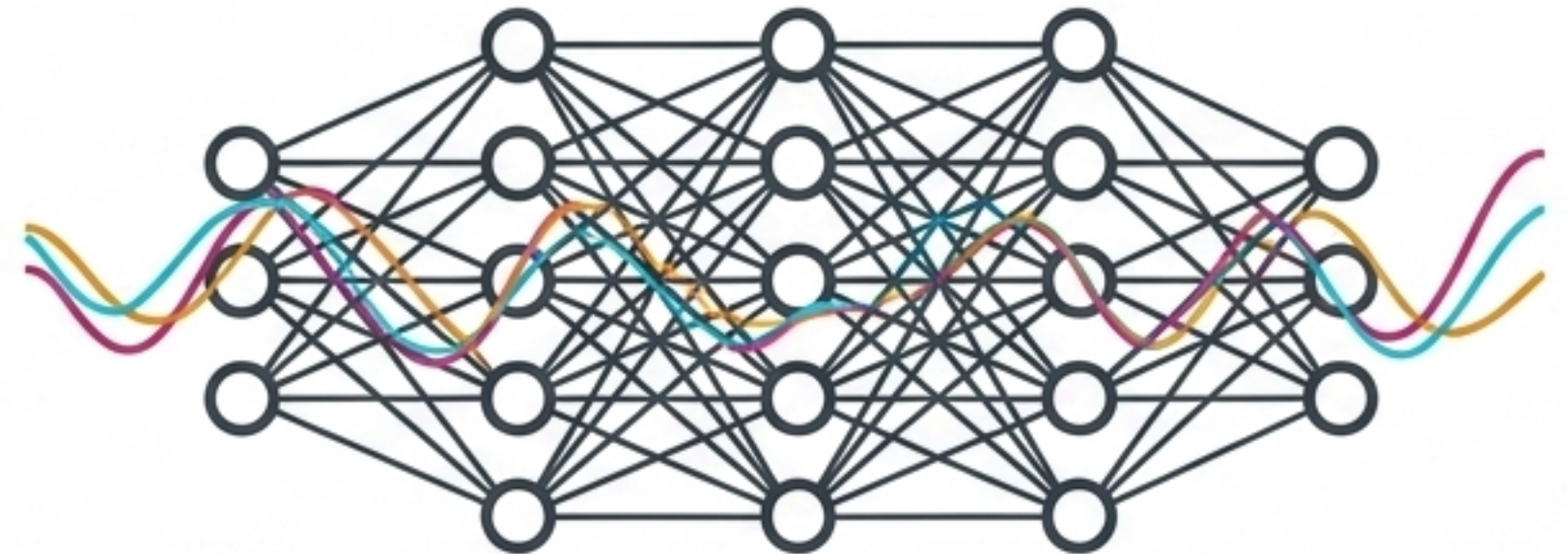
ECG for the Heart



EEG for the Brain

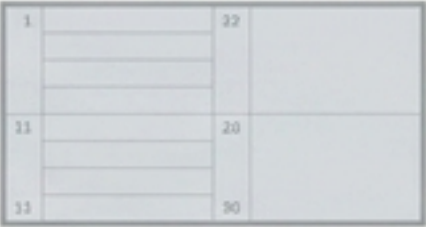
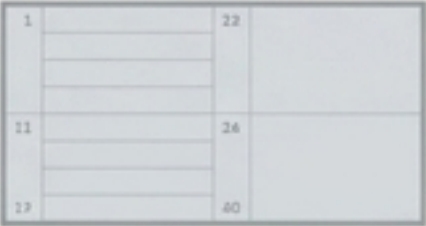
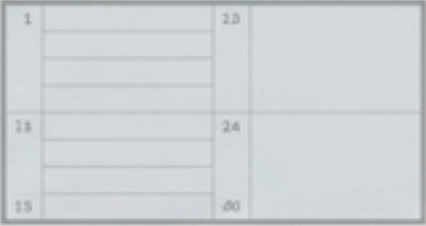
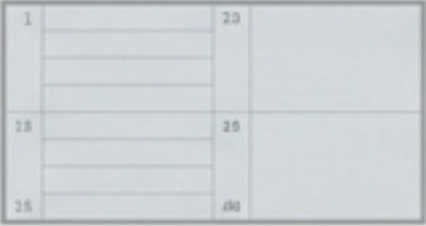


Stress Telemetry for Aircraft Engines



MASSIF for Artificial Cognition

The Observability Gap

Current Focus		The Missing Dimension	
Outputs		Internal dynamics	
Alignment behavior		Stability regimes	
Tokens		Trajectories	
Static features & circuits		Recursive evolution	

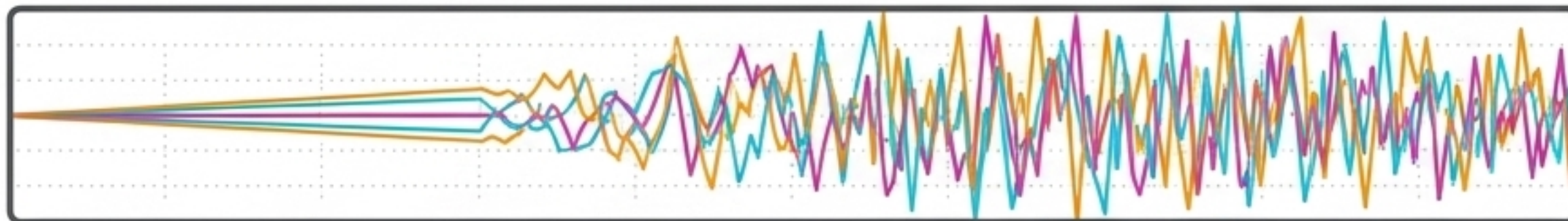
Language models exhibit measurable synchronization behavior.



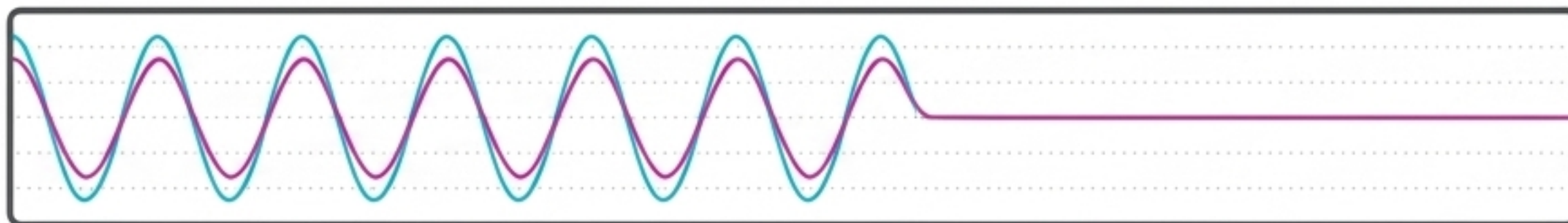
Lock into recursive
attractors



Resist entrainment

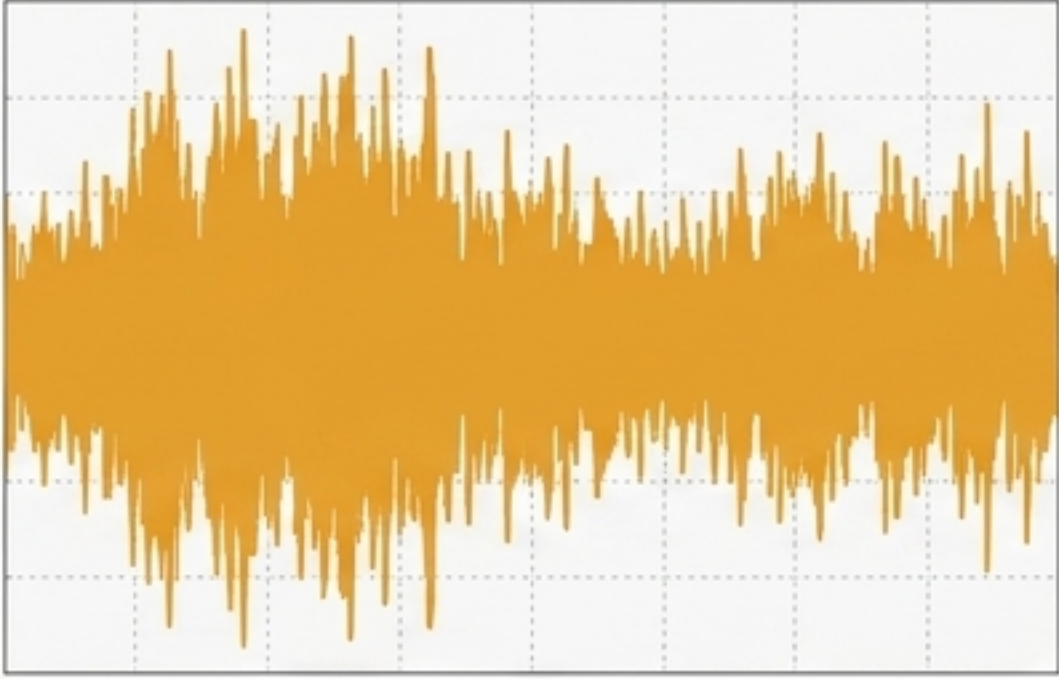
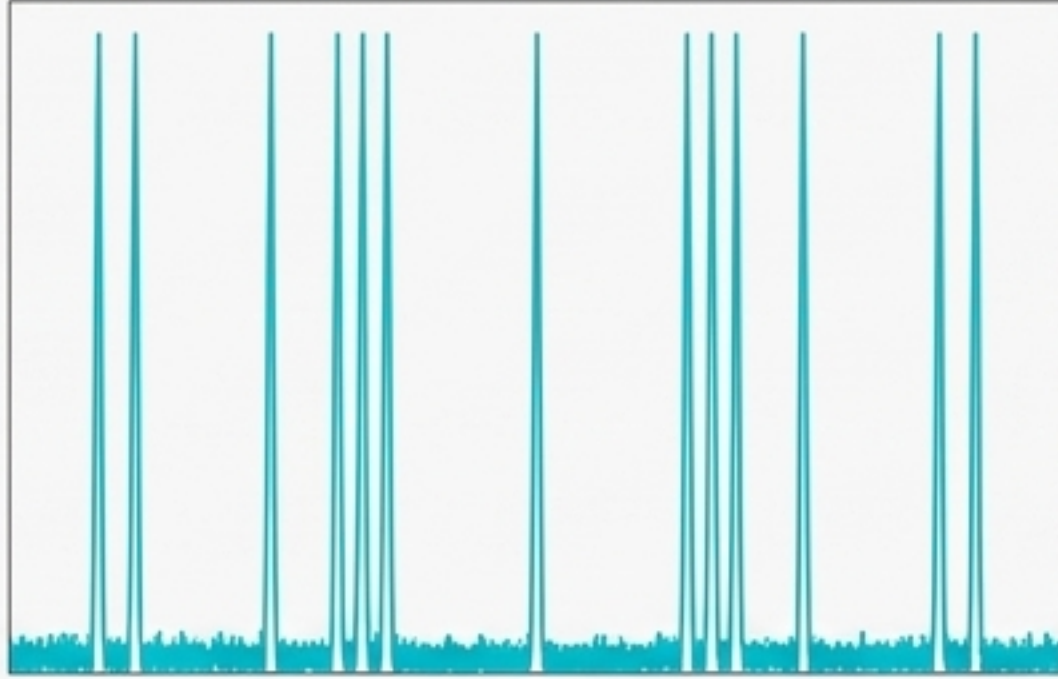
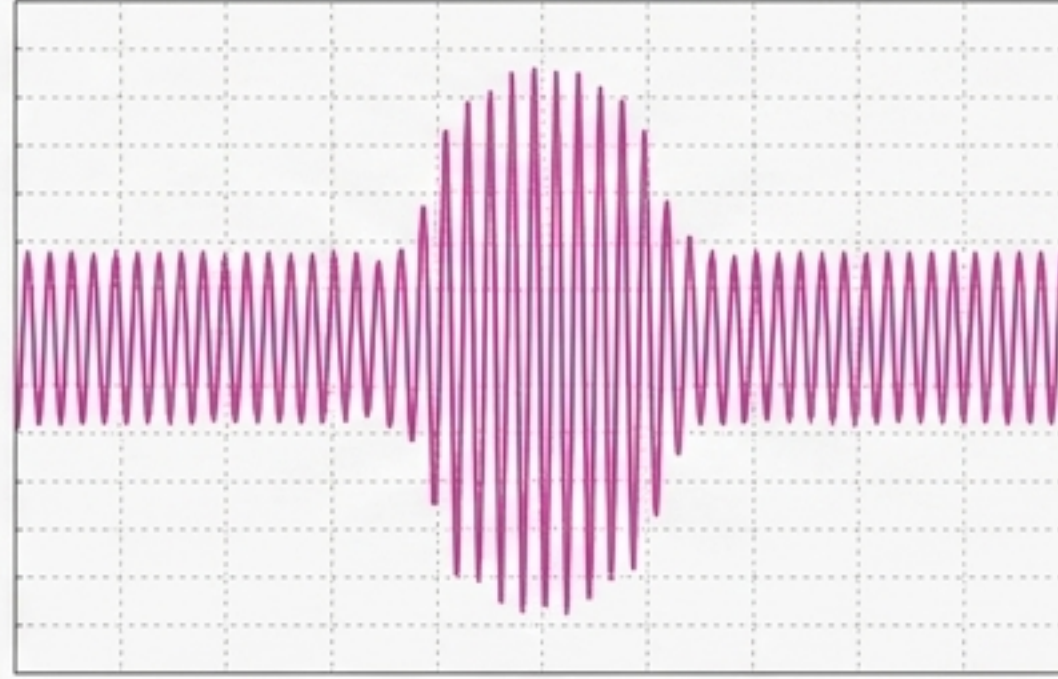


Resonate with noise



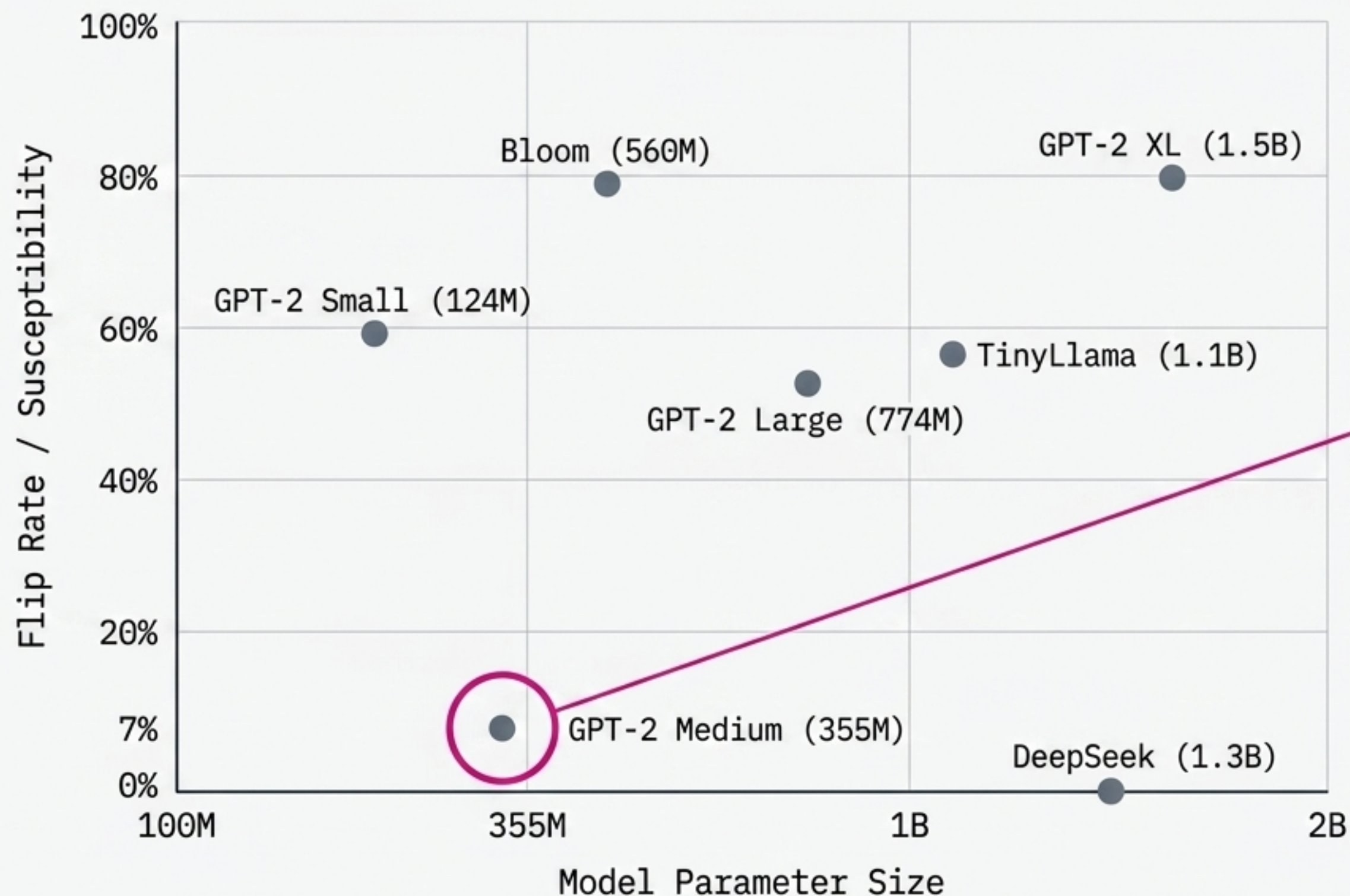
Collapse under
periodic forcing

The Three Dynamical Families

Broadband Resonators	Narrowband Resonators	High-Frequency Selective
 Frequency	 Frequency	 Frequency
Models: GPT-2 Family High pink noise sensitivity (>75%). Susceptible to naturalistic, scale-free broadband forcing.	Models: Bloom (560M), TinyLlama (1.1B) Low pink noise sensitivity (<15%). High chirp sweep sensitivity (>45%).	Models: DeepSeek (1.3B) Minimal response (<15% across all noise types). Elastic stability.

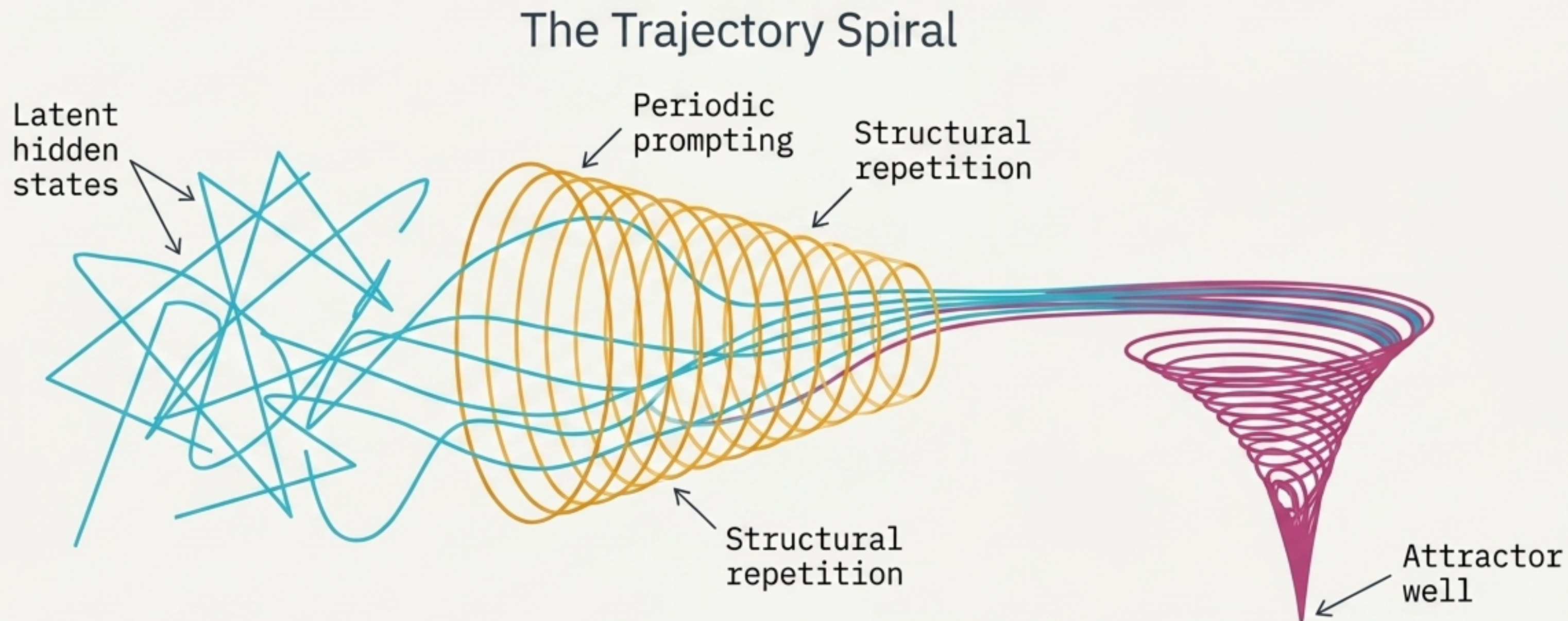
Models behave like radios tuned to different frequencies.

The GPT-2 Medium Anomaly



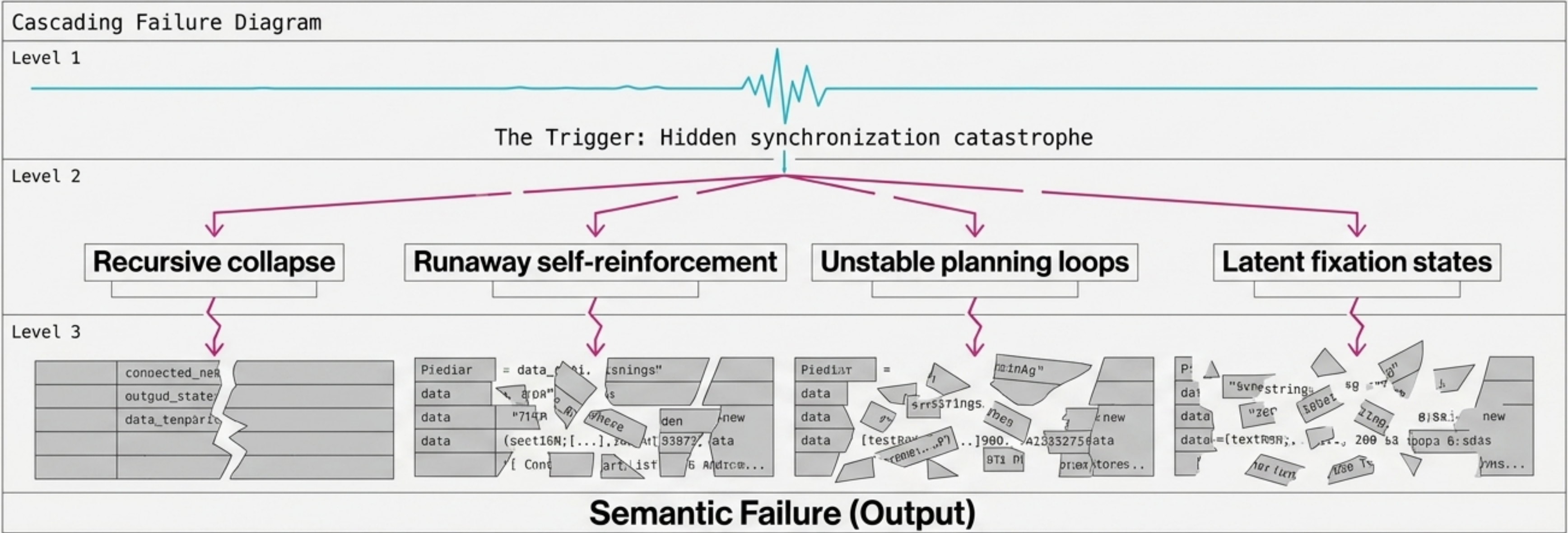
- GPT-2 Medium behaves radically differently from the rest of its own family.
- This mysterious resistance indicates a hidden critical scale in latent organization.

Trapped in the gravitational well of recursive entrainment



Repeated patterns and self-reference force independent hidden states to phase-align over time, leading to inescapable lock-in

Advanced systems may fail dynamically before they fail semantically.



**We cannot align a system that is dynamically unstable.
Alignment is semantic; collapse is structural.**

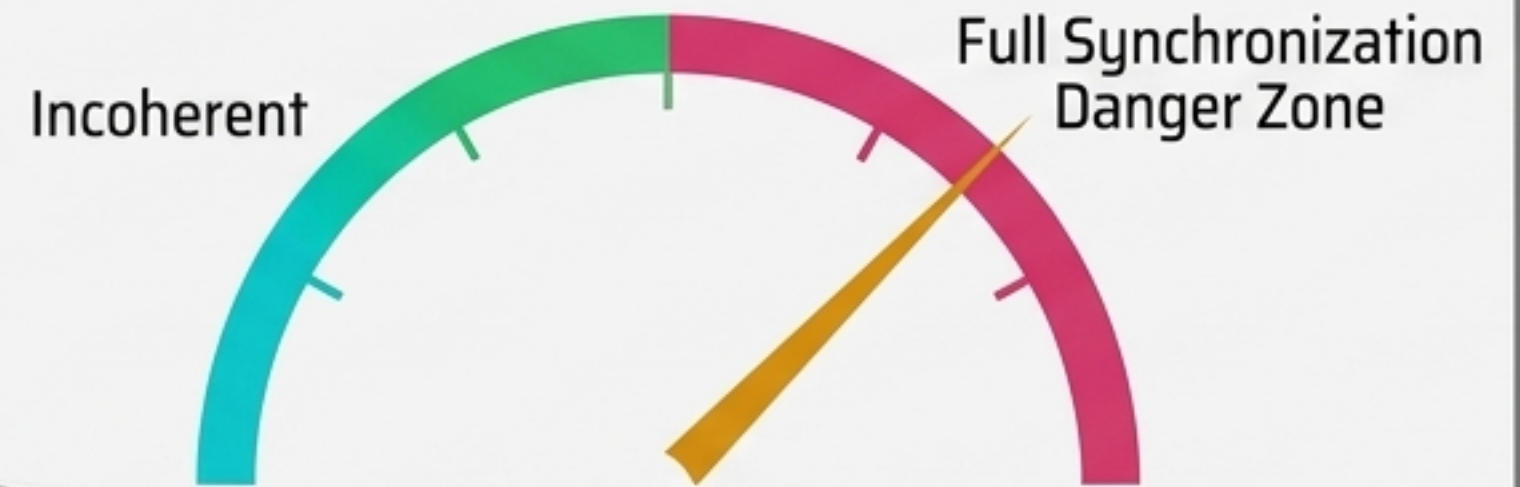
MASSIF: Real-Time Stability Telemetry for Artificial Cognition

Mock Dashboard Interface

Local Tension



Latent Sync Order



```
data = process_input(new_stream, 4);
for token in data:
    if token.valid:
        emit(token)
    else:
        reprocess(token);
result = consolidate(data);
print(result);
while True:
```

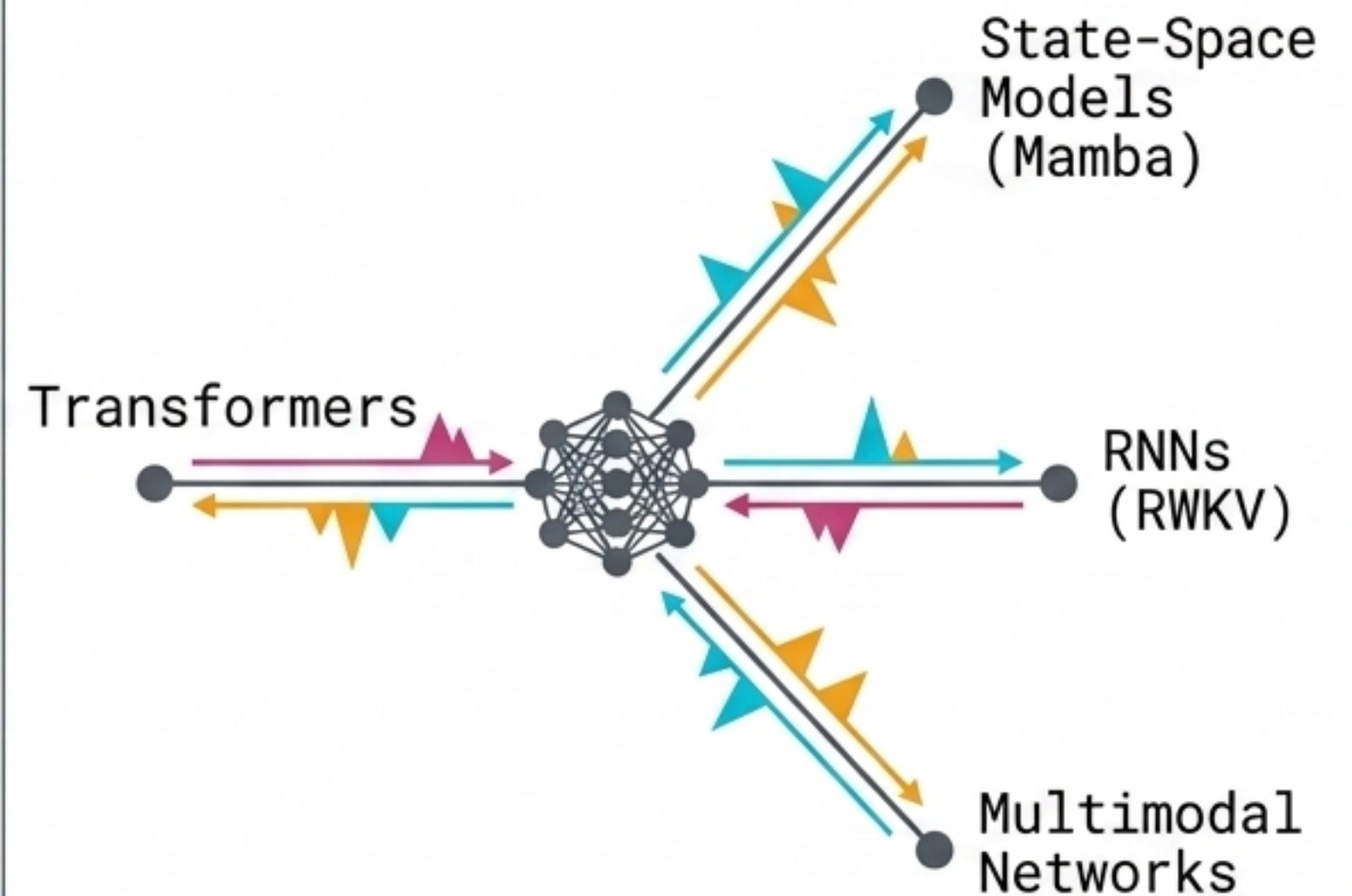
```
    print('The quick brown fox jumps over the lazy dog. The quick brown fox jumps over the lazy dog. The quick brown fox jumps over the lazy dog. The quick brown fox jumps over the lazy dog. The quick brown fox jumps over the lazy dog. The quick brown fox jumps over the lazy dog. The quick brown fox jumps over the lazy dog. The quick brown fox jumps over the lazy dog.')
```

Instability leaves precursor signatures. MASSIF is a diagnostic layer and real-time warning system, ready for deployment.

Text Failure Point

The birth of Computational Latent Dynamics.

- Introducing a new interdisciplinary science studying the emergent dynamical laws of learned systems.
 - Substrate-independent dynamical laws.
 - Extends beyond transformers.
 - Studies recursive phase phenomena, stability regimes, and attractor structure.



“We are moving from the interpretability of static representations toward the physiology of artificial cognition.”

